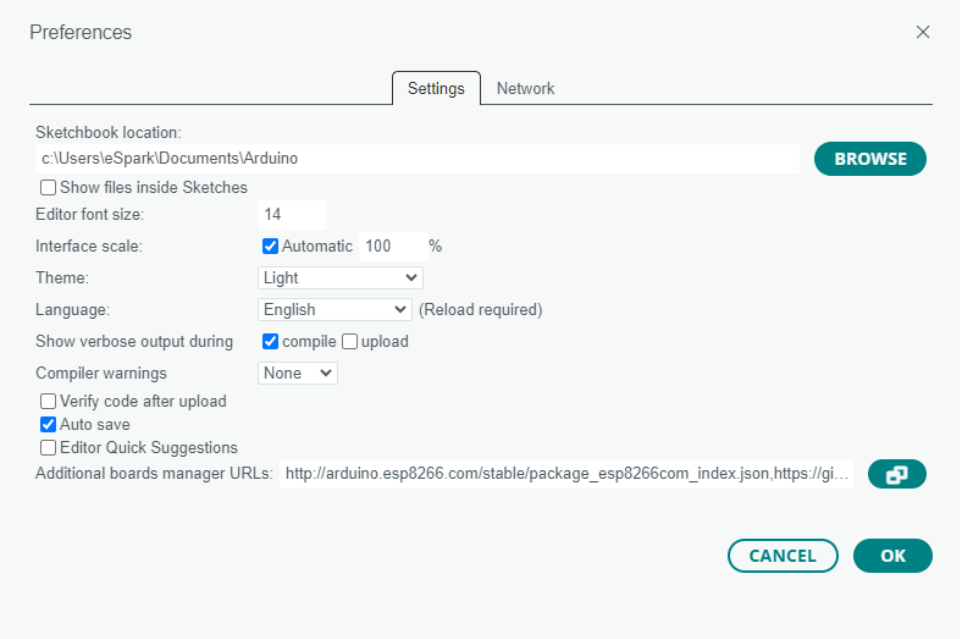
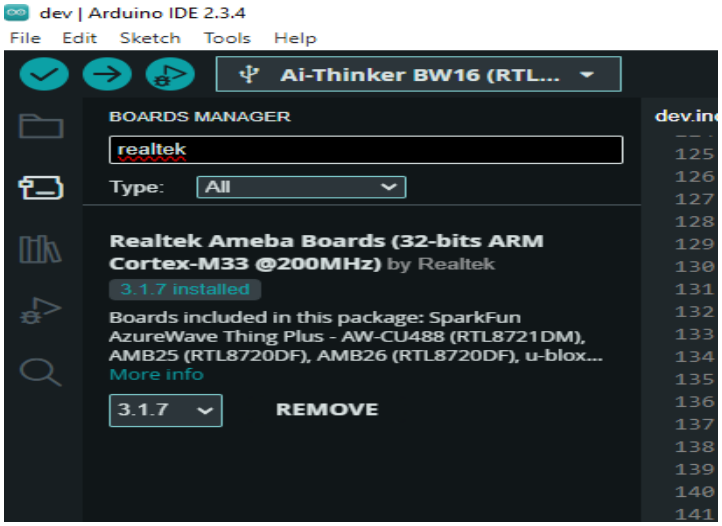
FLASHING AND MONITORING INSTRUCTIONS FOR Real-tek BW16 (Main controller board)

USING VOLT-LINK and ARDUINO IDE: (recommended)

1. Download and install Arduino IDE, go to file->preferences.



2. Add the following URL in the additional boards manager: https://sandeepmistry.github.io/arduino-nRF5/package_nRF5_boards_index.json
3. Go to the tools->boards->boards manager and install the following board:



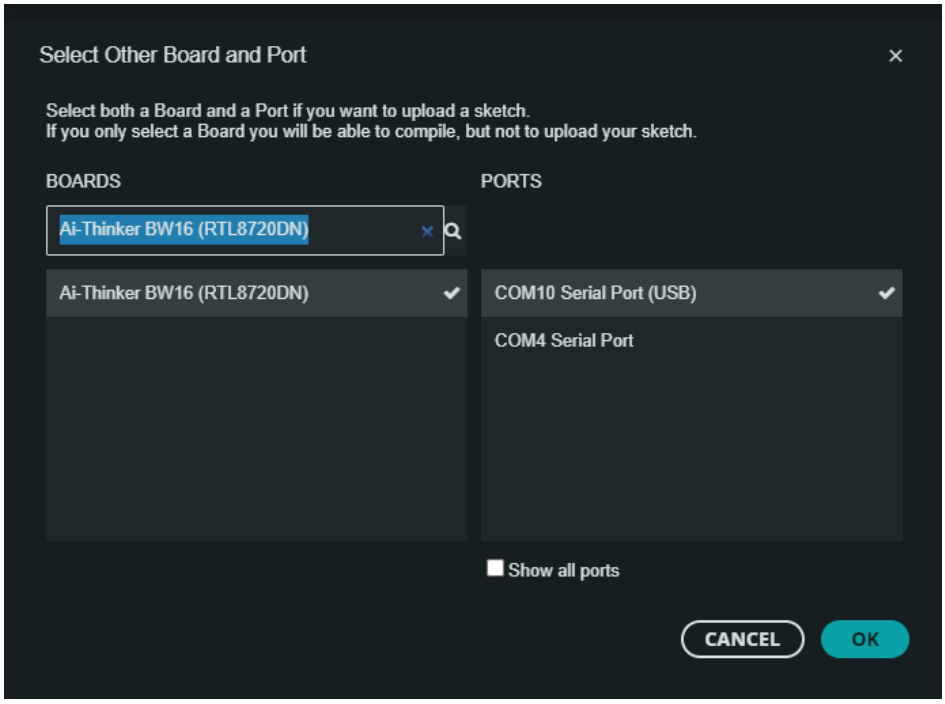
4. Now connect your volt-link to your pc via type c cable. And connect the volt link to the main controller boards J3 port using the connector wire :



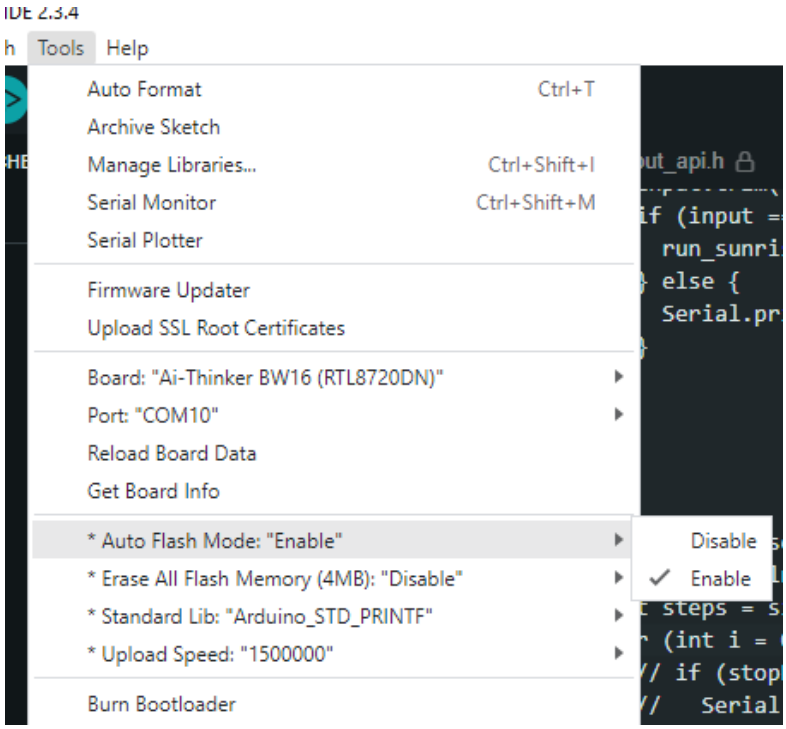
5. Make sure you have CP210x driver installed , if not download using this link: https://www.silabs.com/documents/public/software/CP210x_Universal_Windows_Driver.zip

6. Now go to device manager->ports and see if the device is detected , if its already there then its okay if not then update driver and select the CP210x.zip file downloaded using the link previously.

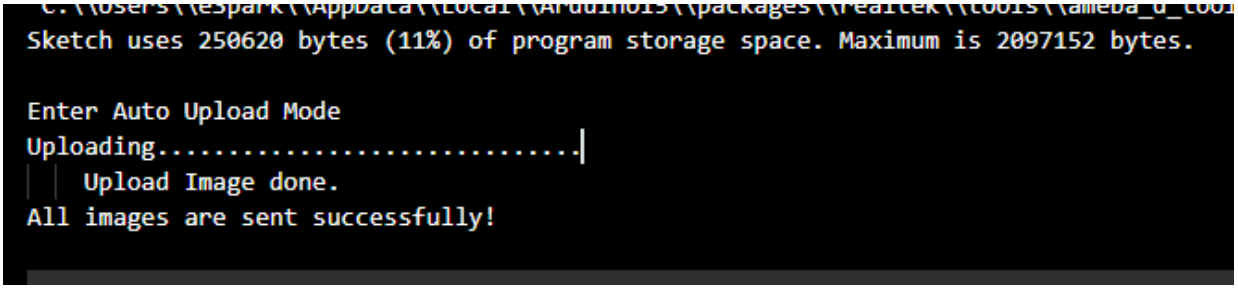
7. Now feel free to upload the source code to a New Sketch in Arduino IDE and make sure to select the correct COM PORT and the correct board:



8. Make sure that the auto flash mode is enabled:

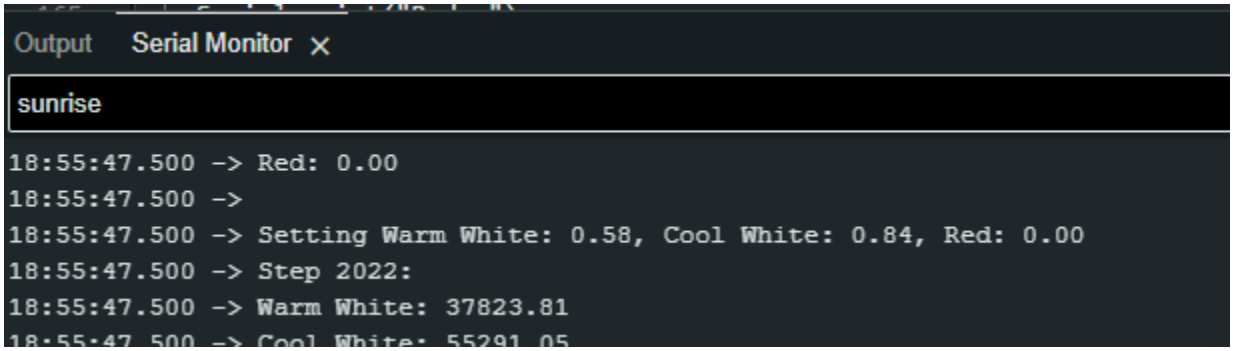


9. Compile and upload the **code (make sure the board is given supply i.e. 24V else code will not be uploaded)**. You will observe the following logs on a successful flash:



10. To monitor logs open a serial monitor in the Arduino ide by navigating to tools->serial monitor , set the baud rate to 115200

11. In order for the sunrise animation to begin give a serial command using the serial monitor , the command is “sunrise”. After pressing enter the animation should begin , which will last upto 60 seconds approx..



Following logs should be observed after running the animation:

```
18:55:47.843 -> Starting sunrise sequence...
19:35:17.248 -> Setting Warm White: 0.00, Cool White: 0.00, Red: 0.00
19:35:17.248 -> Step 1:
19:35:17.248 -> Warm White: 0.00
19:35:17.248 -> Cool White: 0.00
19:35:17.248 -> Red: 0.00
19:35:17.248 ->
19:35:17.248 -> Setting Warm White: 0.00, Cool White: 0.00, Red: 0.00
19:35:17.292 -> Step 2:
19:35:17.292 -> Warm White: 0.00
19:35:17.292 -> Cool White: 0.00
```